Criptografía, claves y direcciones

Criptografía

- Criptografía significa escritura secreta en griego
- Pero la ciencia abarca mucho más
  - Confirmar un mensaje secreto sin revelarlo (firma digital)
  - Verificar la autoría (fingerprint)

Problemas de enviar mensajes seguros

Criptografía Simétrica

Criptografía asimétrica

Caesar Cipher

Claves

- Las claves no se almacenan en la red sino en wallets.
- Las claves consisten en dos claves realmente:
  - Pública (bank account number)
  - Privada (similar a PIN)

Public Key Infrastructure

¿Cómo podemos estar seguros que Bob es realmente la clave de Bob y no de otro?. Muy importante para sitios Web.

Varias alternativas: Certificate authorities, Web of trust, SPKI

En muchas Cryptomonedas, no hay un PKI

Not your key, not your money

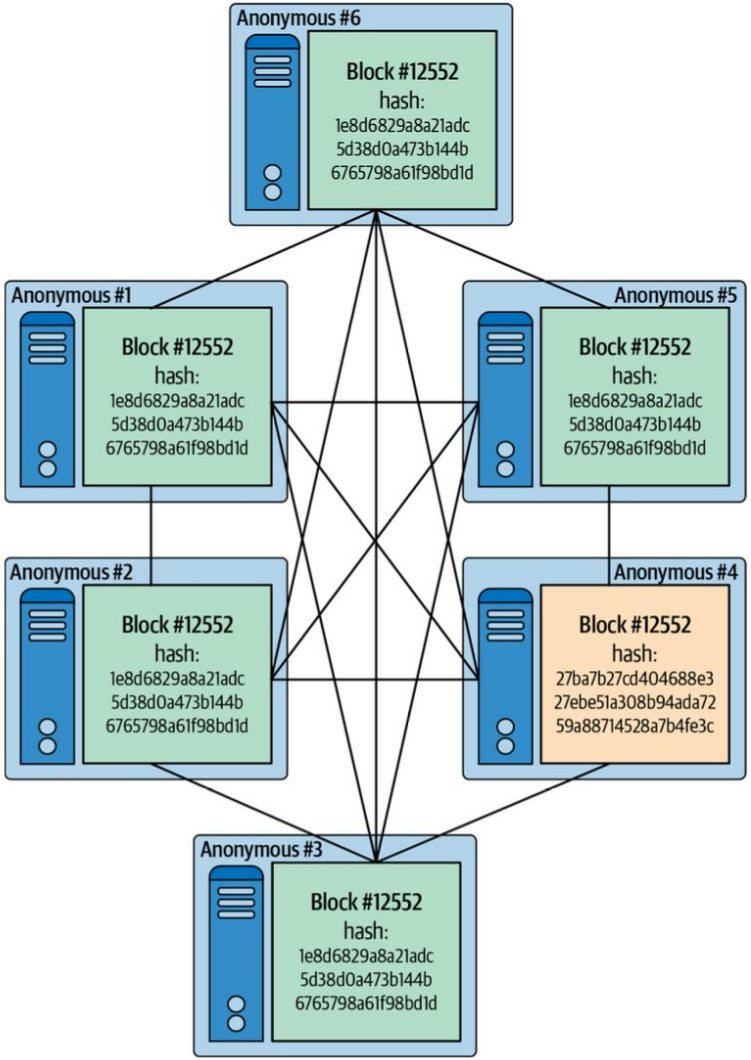
One-way function

- Dado f(x)=y. Es muy difícil resolver x a partir de y pero si tenemos x es posible calcular y.
- Por ejemplo, supongamos la función "al cuadrado" desde un punto de vista humano.
  - f(27)=y es relativamente fácil de obtener 27*27 → 729
  - Sin embargo, la inversas f', x= f'(729) es mucho más difícil

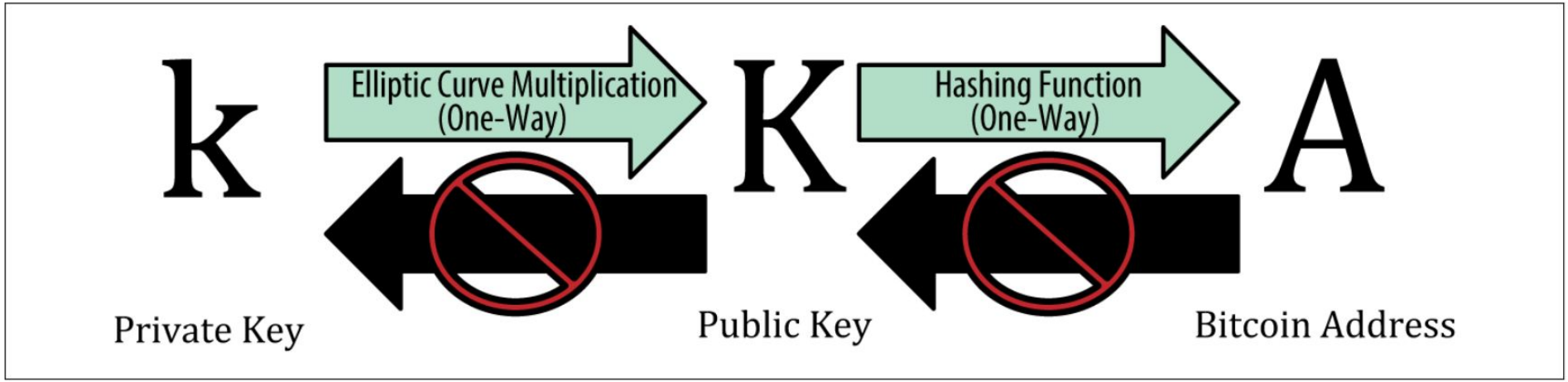
Que es una función hash

- Definición
  - Es una función que permite mapear datos de longitud variable a un valor de tamaño fijo.
  - Determinístico
  - **Tolerante a colisiones (propiedad deseada)**
- Estándares:
  - Bitcoin→ SHA-256
  - Ethereum→ Keccak-256
- Ejemplo de función:
  1. Convertir una secuencia de caracteres a su valor ASCII
  2. Sumar todos los valores
  3. Retornar un módulo de 256 (0x0 - 0xFF)

Un uso de la función en Bitcoin



Clave privada, pública,y direcciones



Clave Privada

La clave privada es un número generado al azar.
Se requiere una semilla, un número entre 1 y 2^{256}

Ejemplo de 256 bits mostrados en 64 dígitos hexadecimales (4 bits)

```
1E99423A4ED27608A15A2616A2B0E9E52CED330AC530EDCC32C8FFC6A526AEDD
```

bitcoind:

```
 $ bitcoin-cli getnewaddress

    1J7mdg5rbQyUHENYdx39WVWK7fsLpEoXZy

$ bitcoin-cli dumpprivkey 1J7mdg5rbQyUHENYdx39WVWK7fsLpEoXZy

    KxFC1jmwwCoACiCAWZ3eXa96mBM6tb3TYzGmf6YwgdGWZgawvrtJ
```

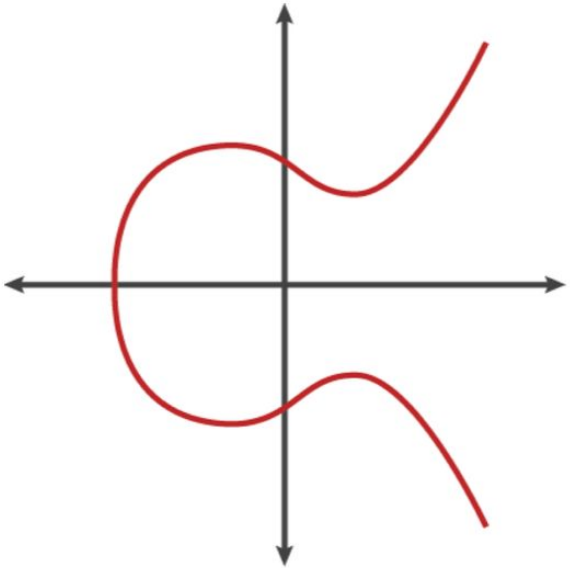
Clave pública

Se basa en Elliptic Curve Cryptography
Sea k = clave privada, G un generador => clave pública es:

K=k*G

- G es siempre el mismo para todos los usuario btc)
- K es un punto en la curva

Bitcoin se basa en secp256k1 definido por National Institute of Standards and Technology (NIST).

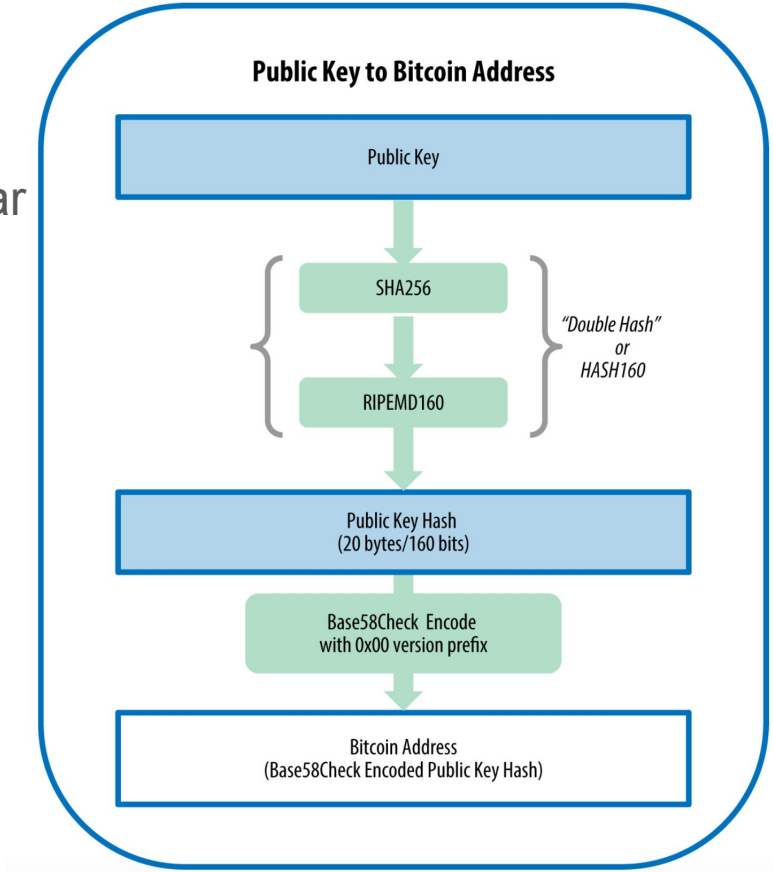


Direcciones

- Es una cadena de caracteres que puede ser compartida con alguien que nos vaya a enviar bitcoin que siempre comienza con 1.

`1J7mdg5rbQyUHENYdx39WVWK7fsLpEoXZy`

- Se obtiene de la clave pública utilizando un algoritmo de hashing de un sentido.



Claves Privadas - Avanzado

- Encrypted Private Keys (BIP-38)

Private Key (WIF)	5J3mBbAH58CpQ3Y5RNJpUKPE62SQ5tfcvU2JpbnkeyhfsYB1Jcn
Passphrase	MyTestPassphrase
Encrypted Key (BIP-38)	6PRTHL6mWa48xSopbU1cKrVjpKbBZxcLRRCdctLJ3z5yxE87MobKoXdTsJ

- Pay-to-Script Hash (P2SH)
- Multisig Addresses

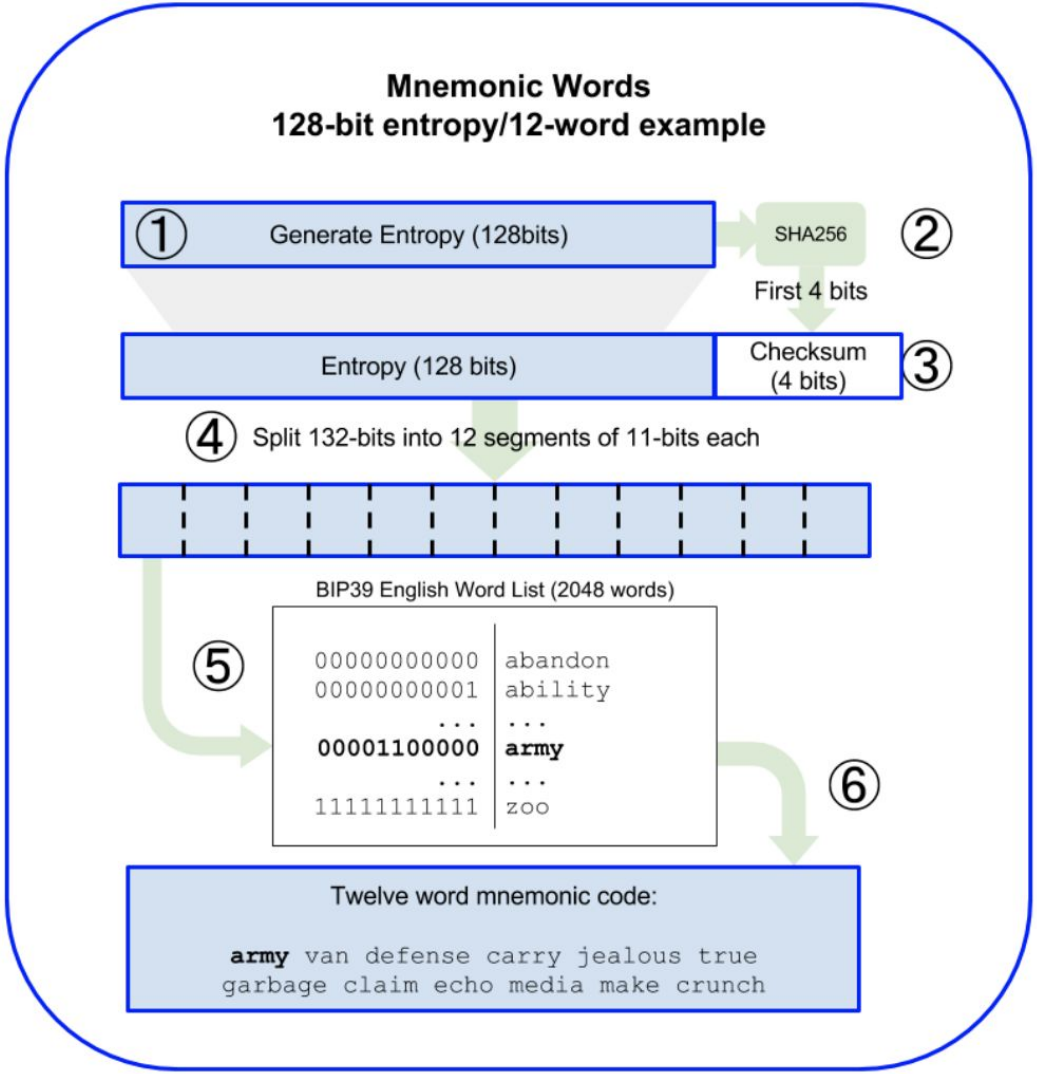
Claves Privadas - Avanzado

- Vanity Addresses
- Paper Wallets

Mejores prácticas

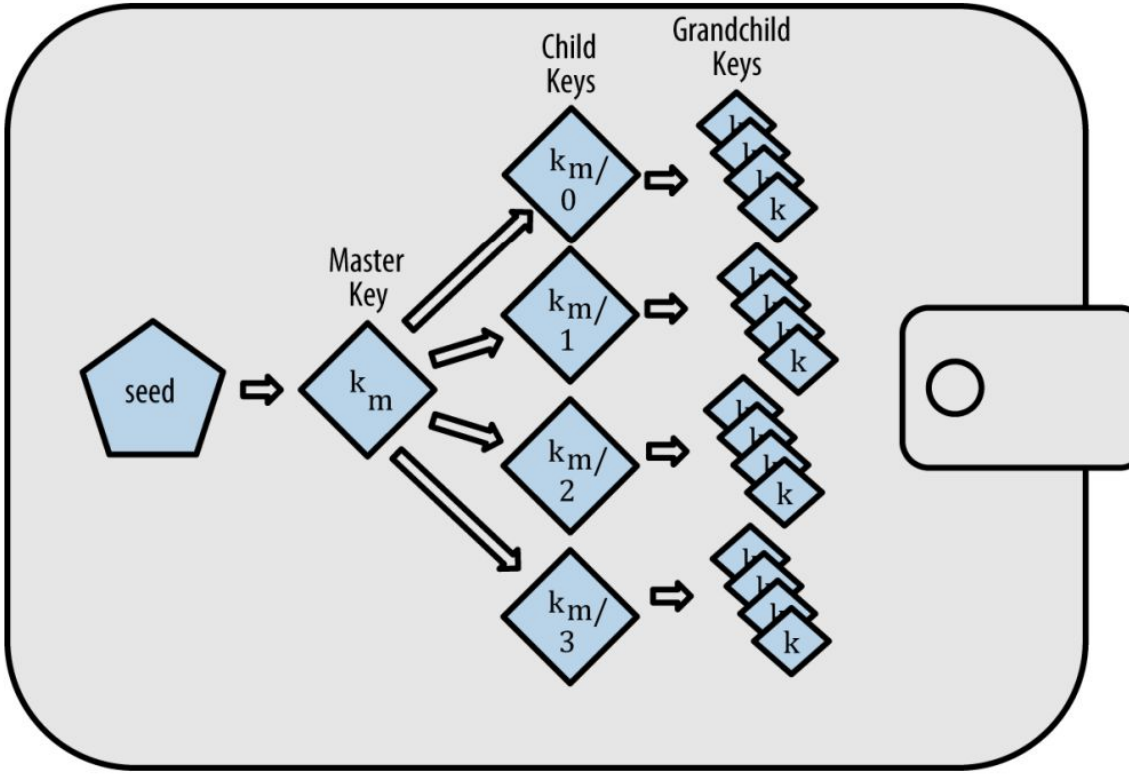
- Mnemonic code words BIP-39

-



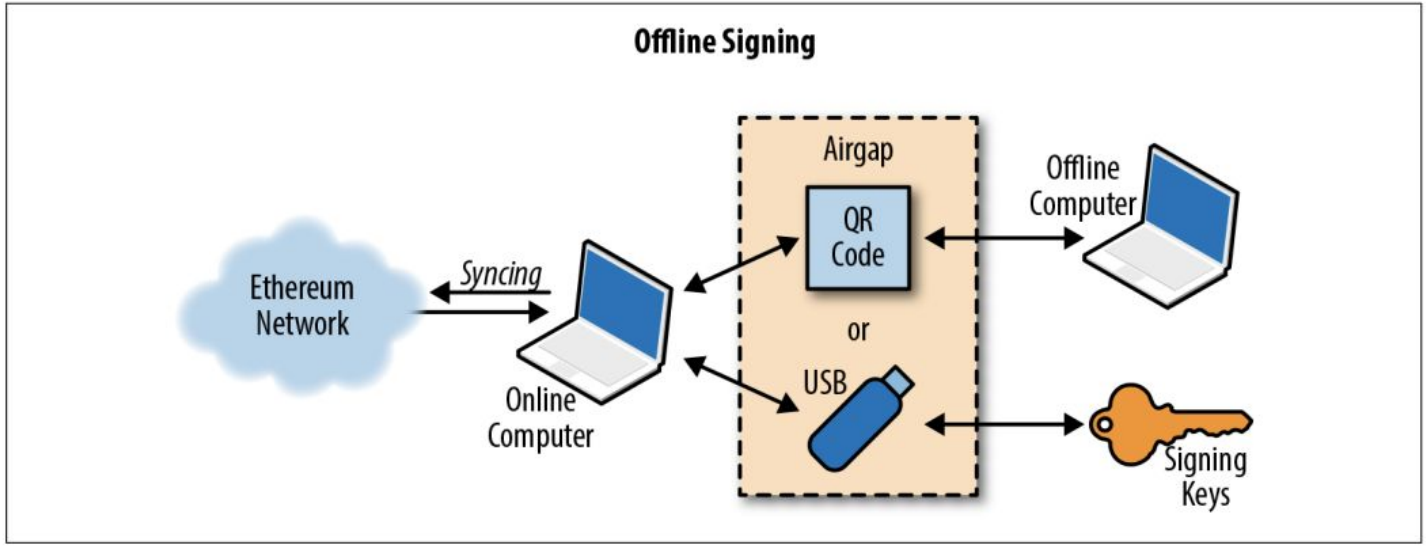
Hierarchical deterministic

- HD wallets BIP-32
- Multipurpose HD wallet structure BIP-43



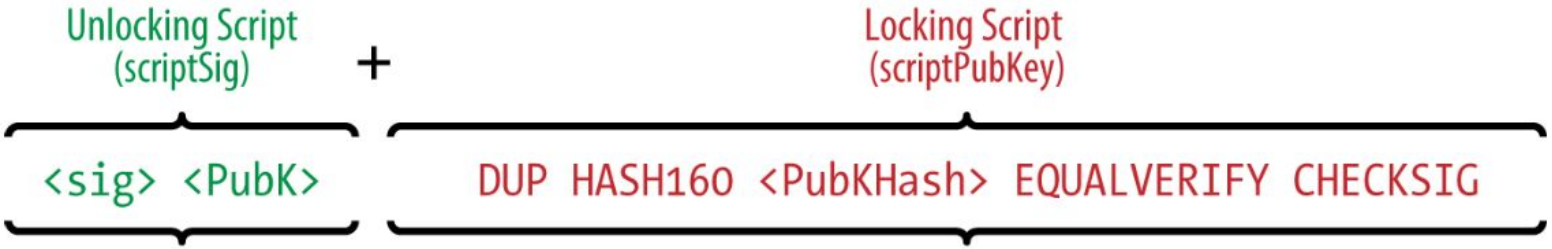
Mejores prácticas

- Offline signing



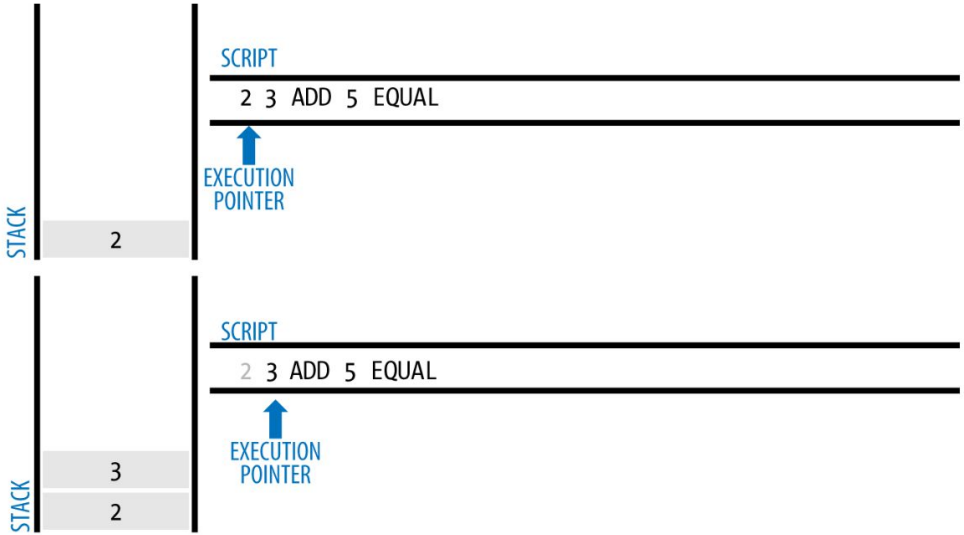
Transaction script

- Turing Incompleteness
  - Especificaciones de complejidad limitada, y predecible
  - No permite implementar bombas lógicas o loops infinitos → ~~DOS~~
- Sin estado
- Utilizado para Lock + Unlock transacciones
  - Condiciones para utilizar una transacción → Output scriptPubKey
  - Resuelve el problema definido en ☝️

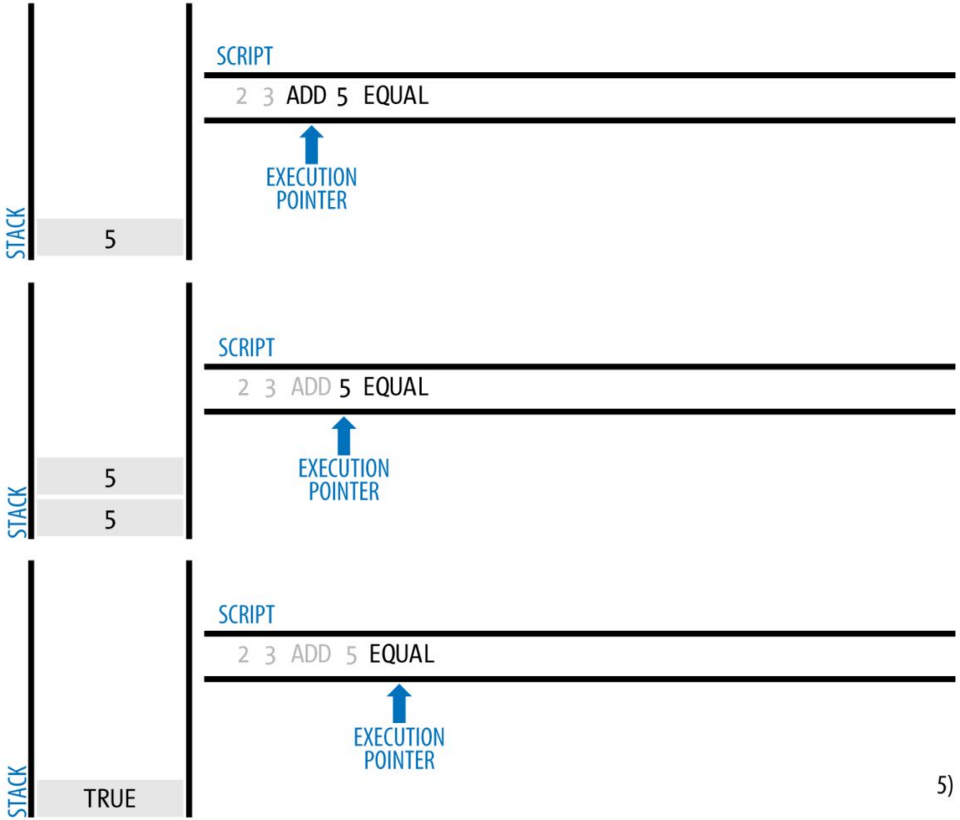


Transaction script

- Se ejecuta en pila
- Ejemplo : 2 3 OP_ADD 5 OP_EQUAL
- Pasos:

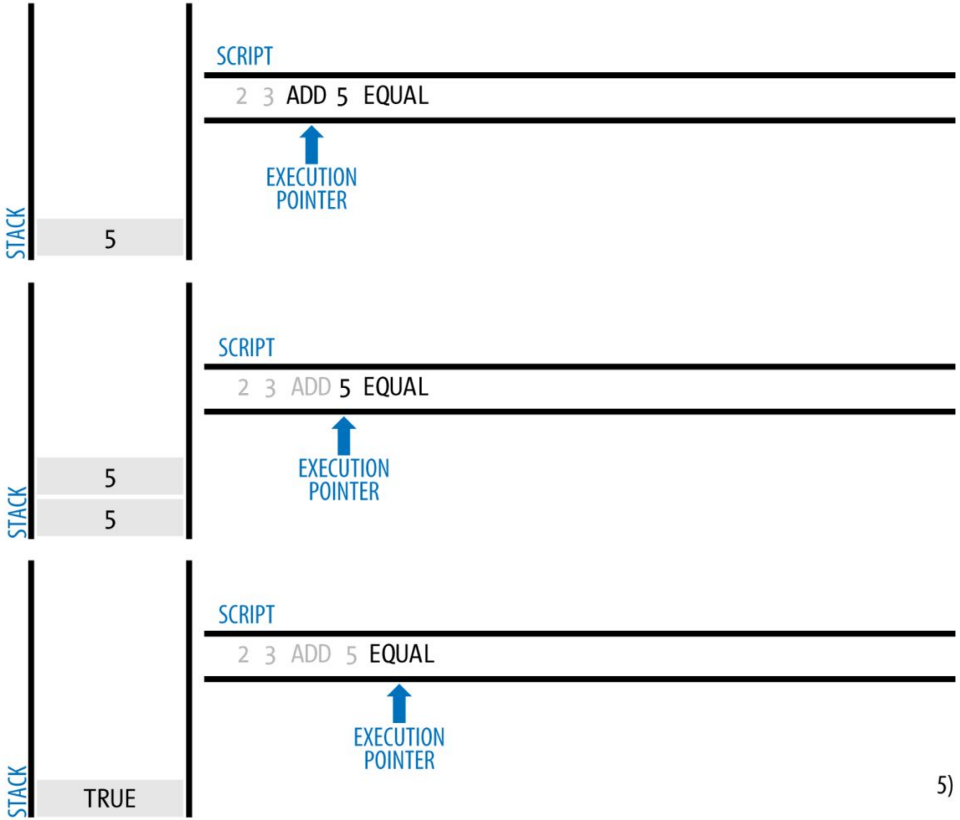


Transaction script



5)

Transaction script



5)

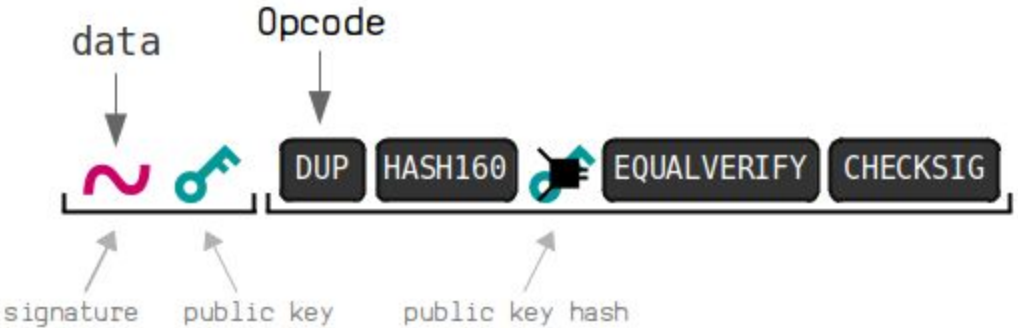
Pay-to-Public-Key-Hash (P2PKH)

- Bitcoin usa P2P para el locking+unlocking
- Bloquea una transacción en base a una clave pública/dirección
- Para desbloquear se necesita una clave pública + una firma obtenida con la clave privada
- Supongamos el caso en que se Alice quiera pagar un café a Bob

```
<Cafe Signature> <Cafe Public Key> OP_DUP OP_HASH160

<Cafe Public Key Hash> OP_EQUALVERIFY OP_CHECKSIG
```

Pay-to-Public-Key-Hash (P2PKH)



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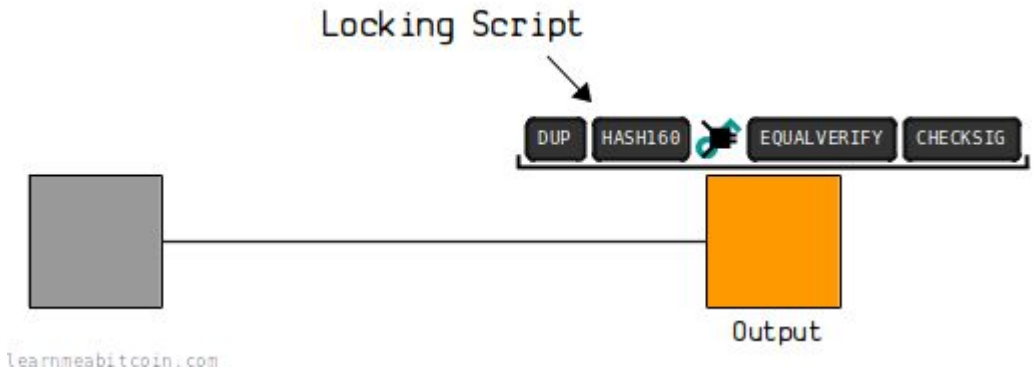

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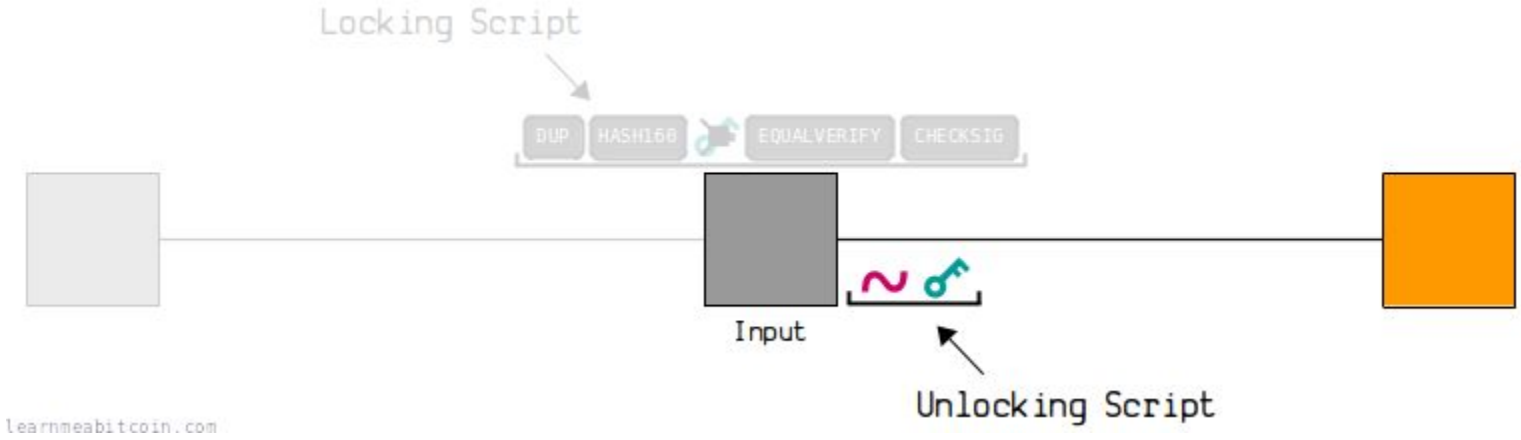

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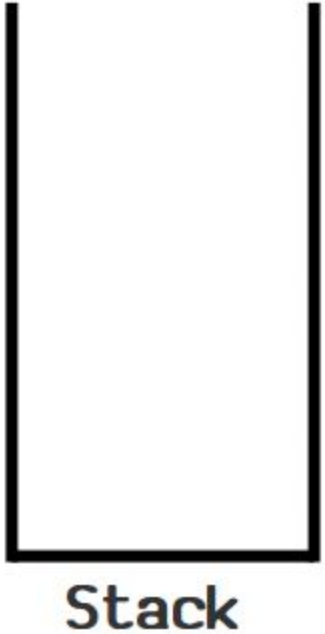

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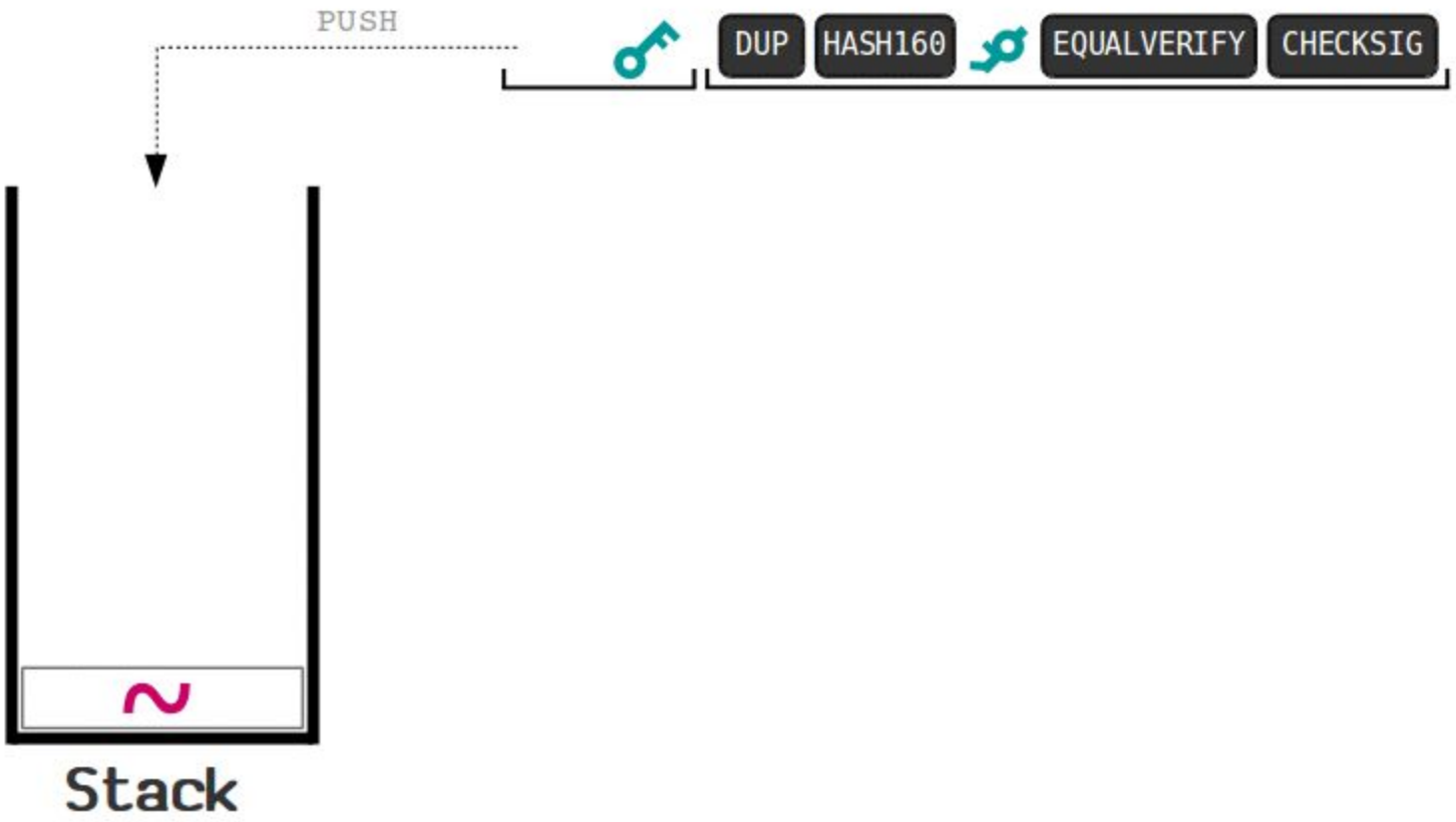

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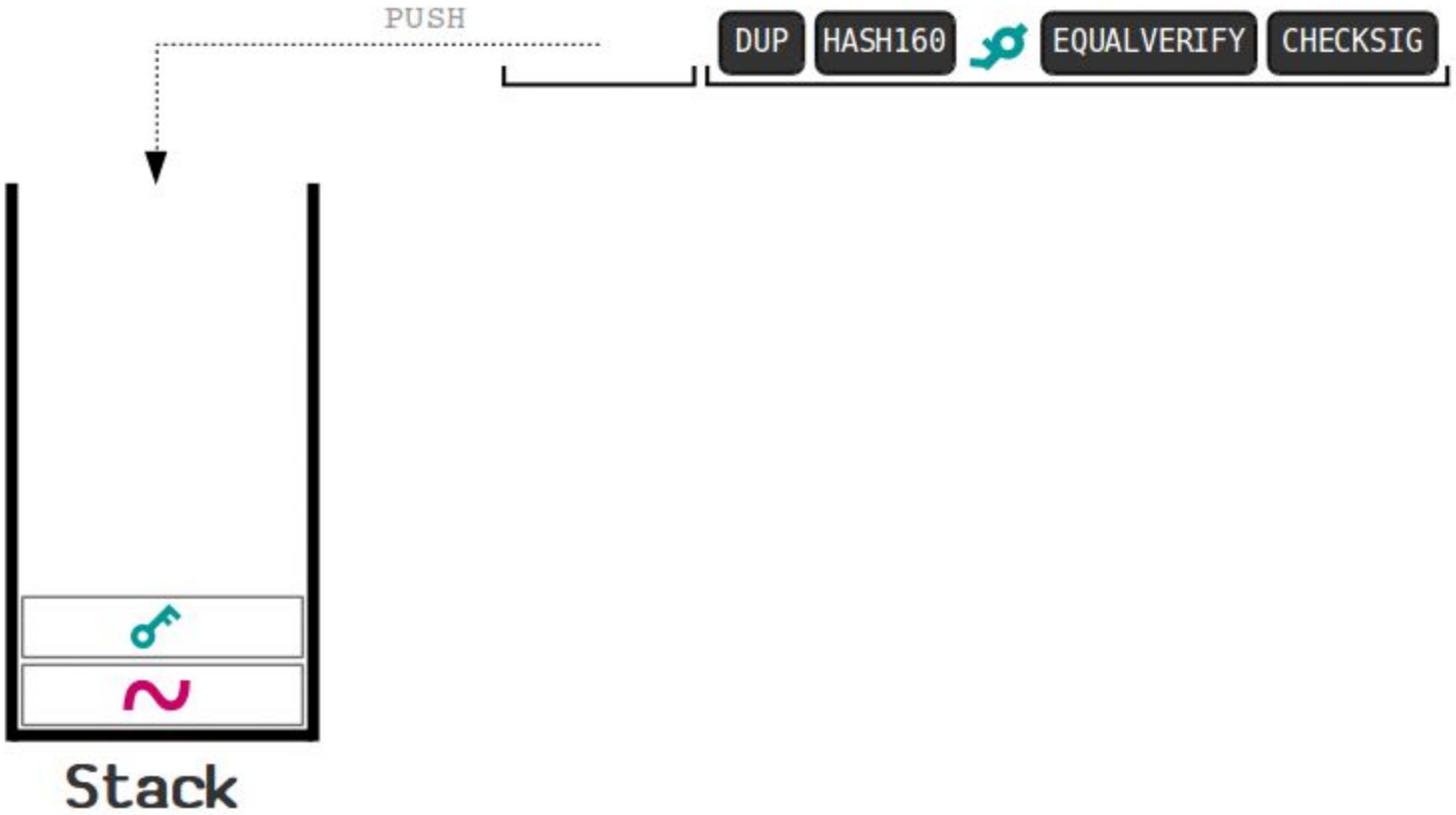

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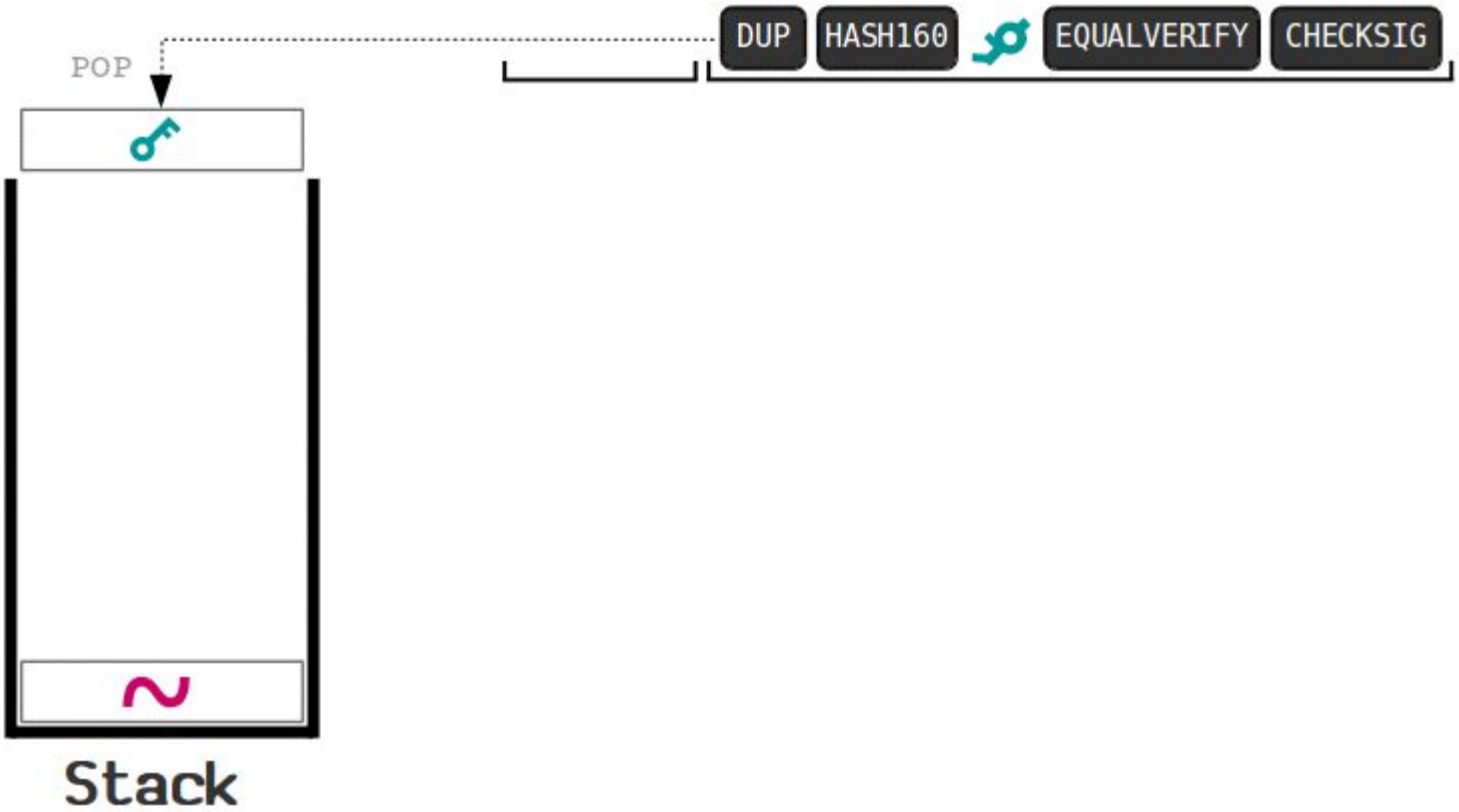

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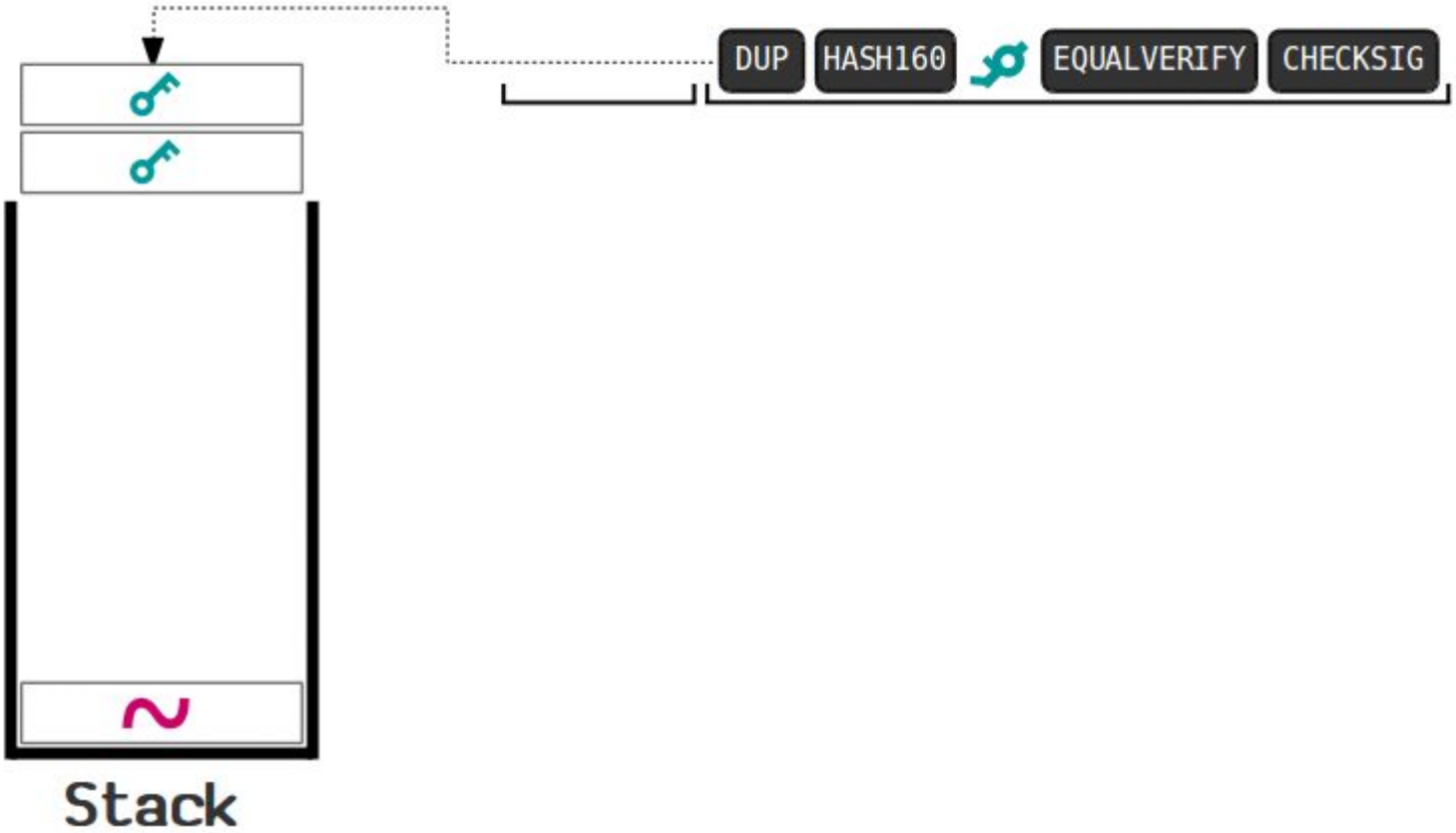

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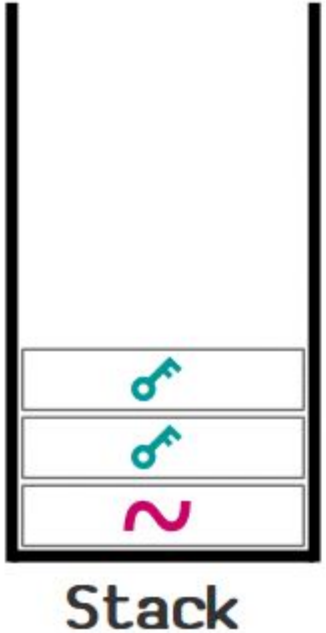

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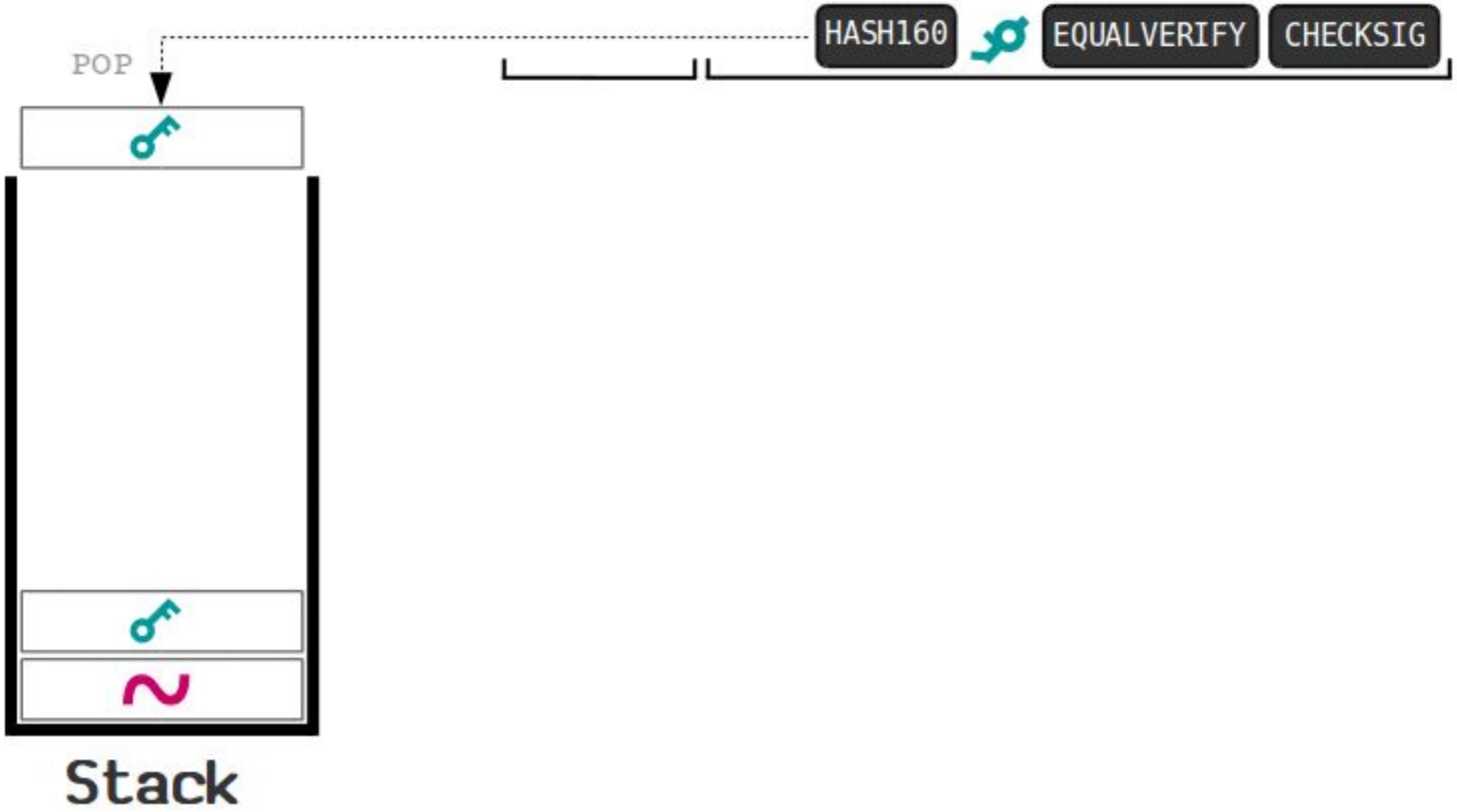

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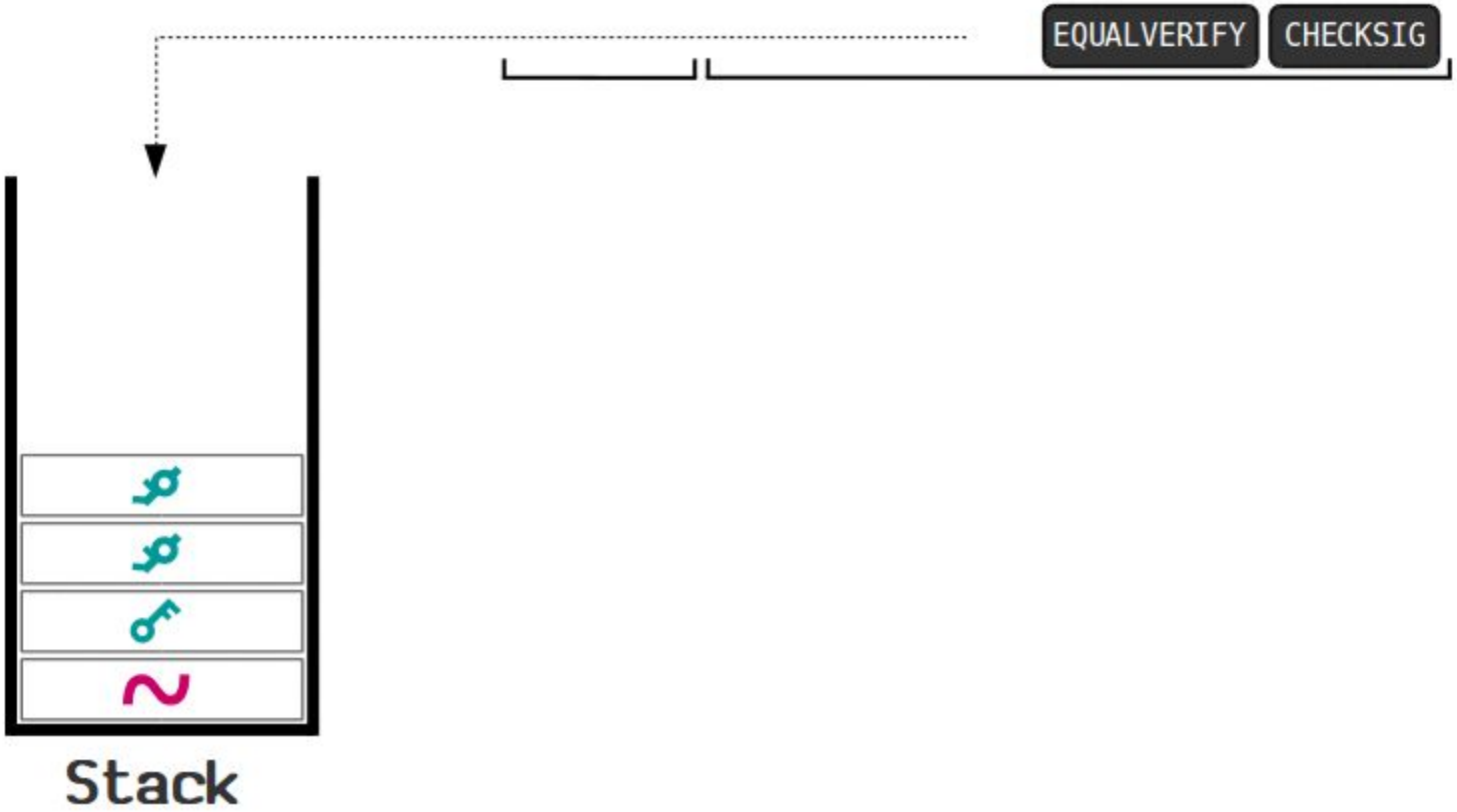

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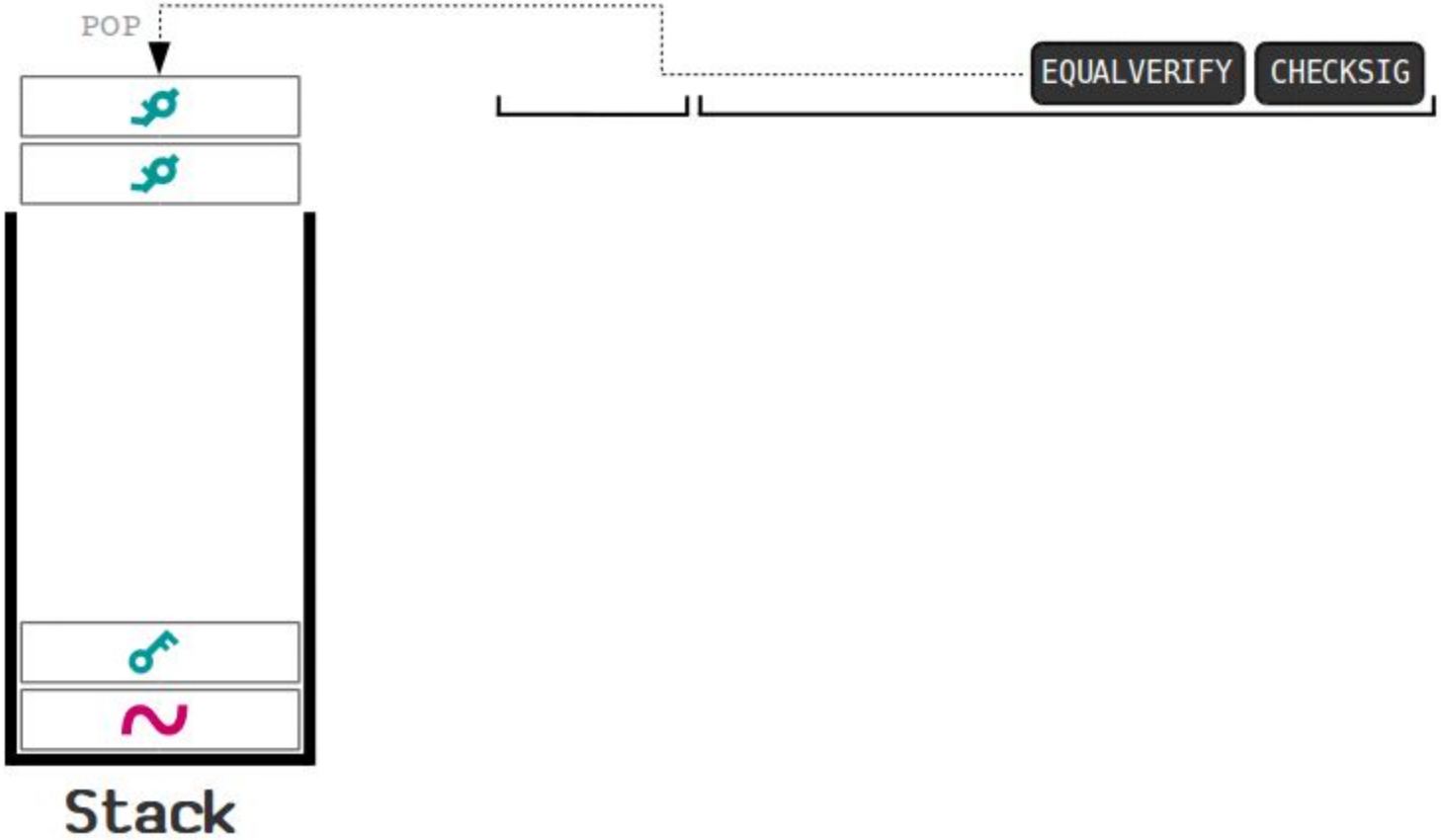

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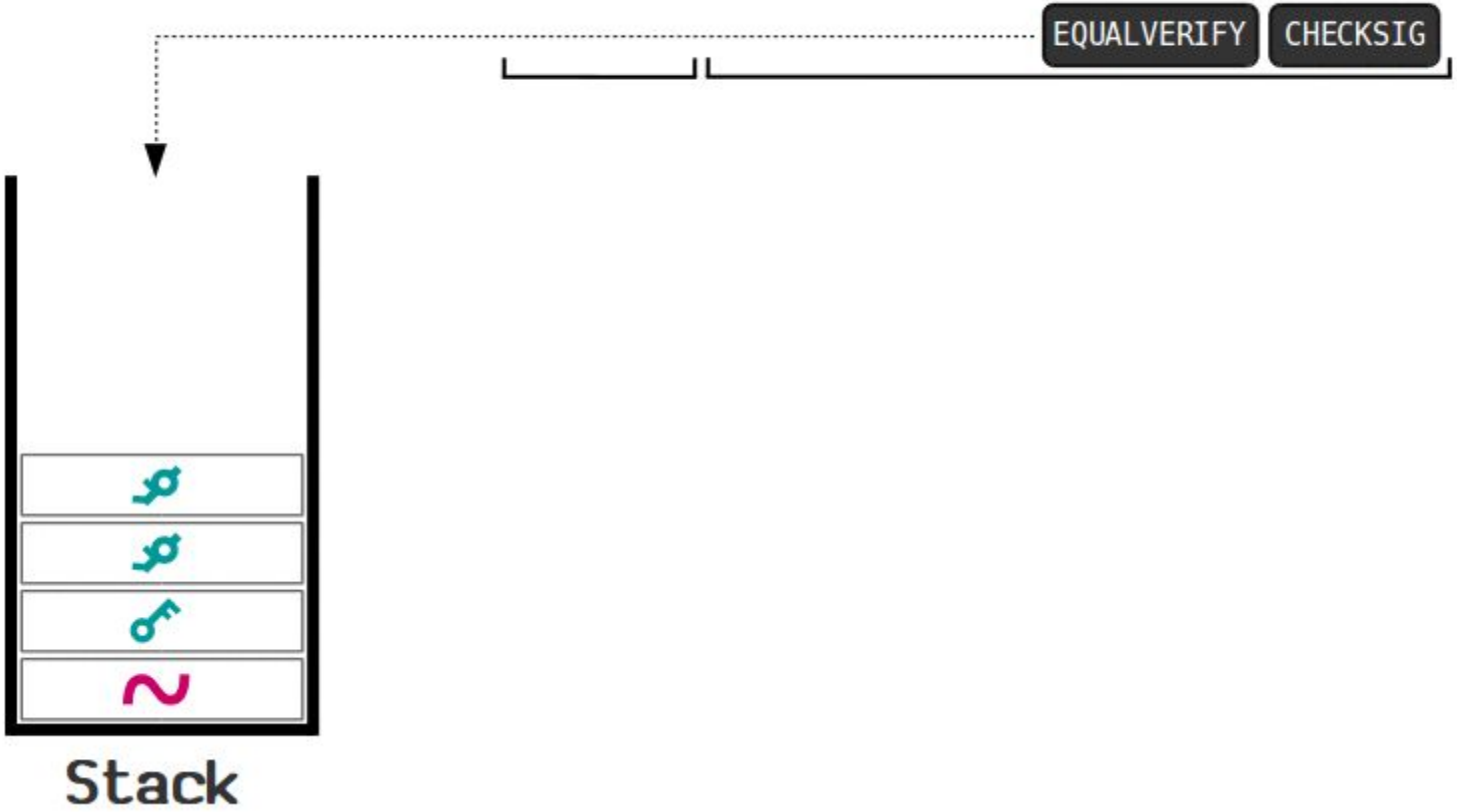

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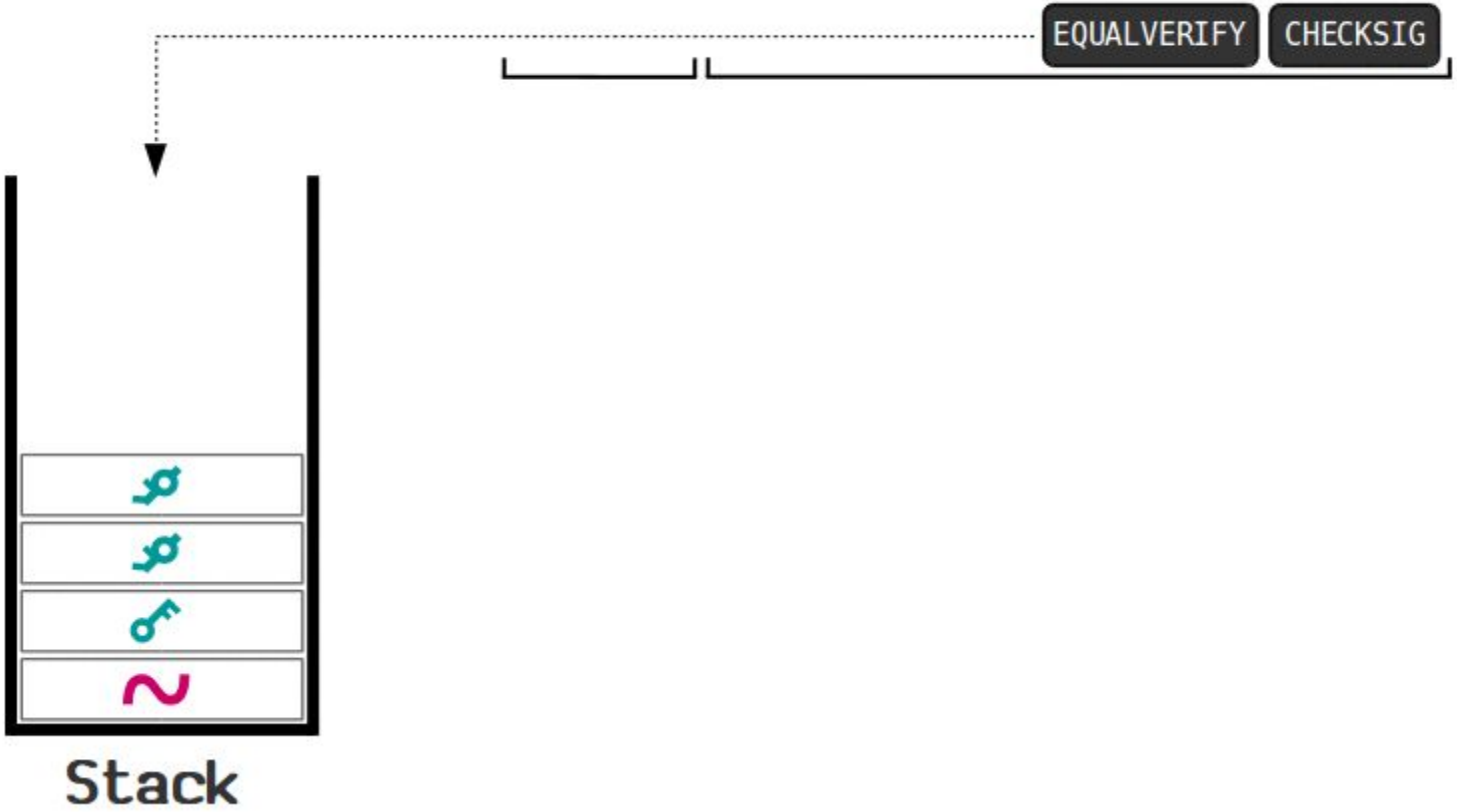

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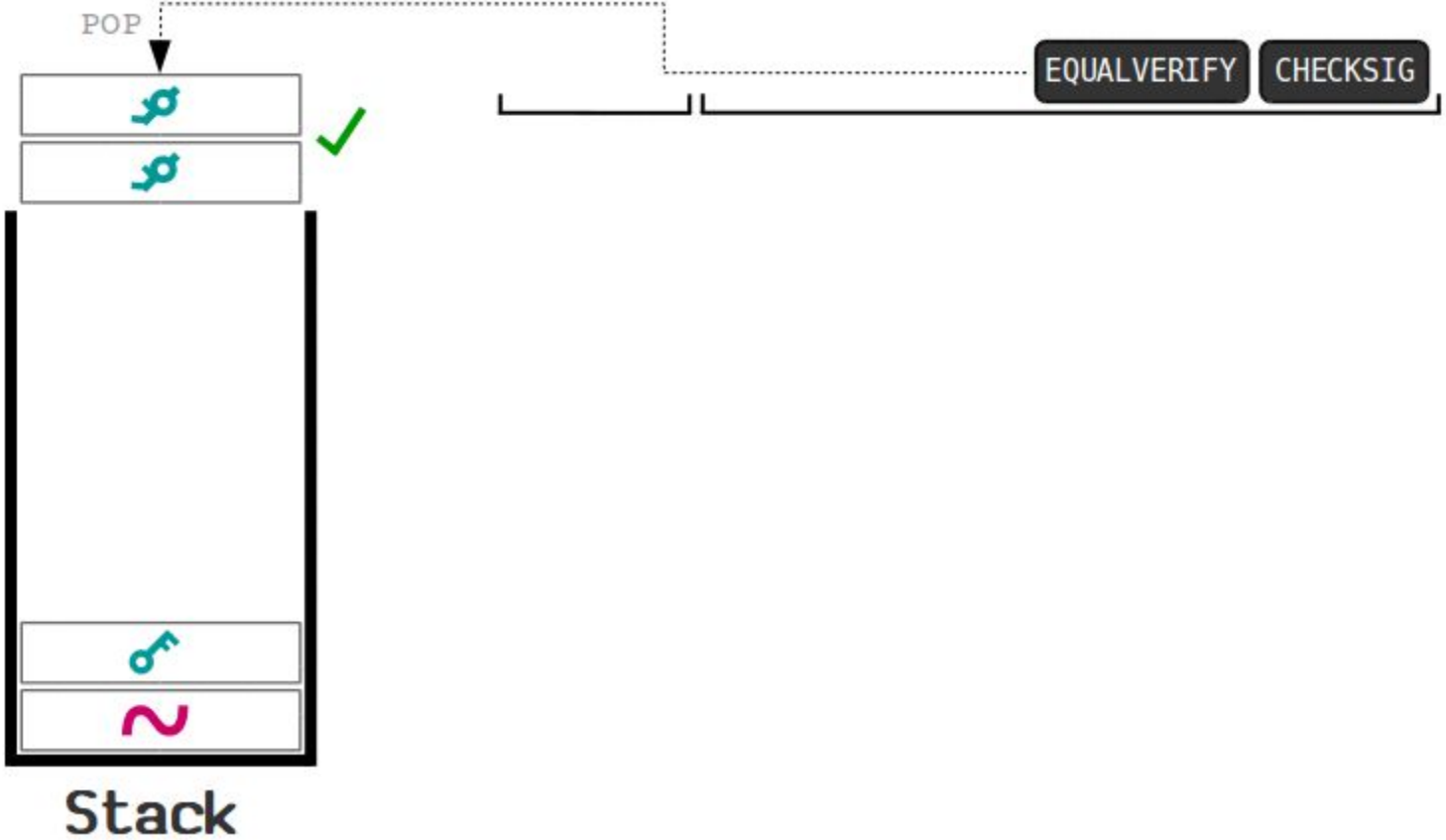

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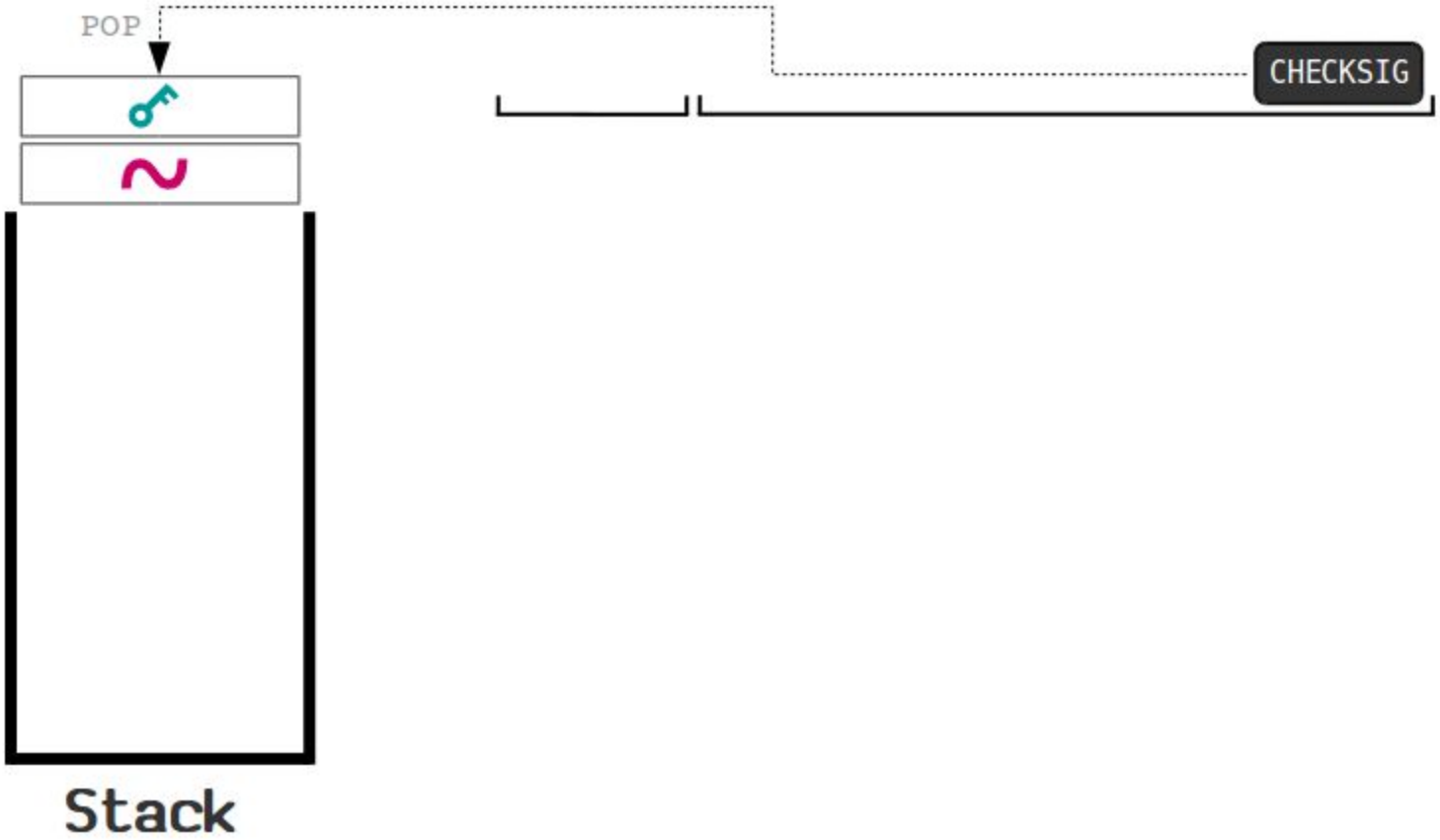

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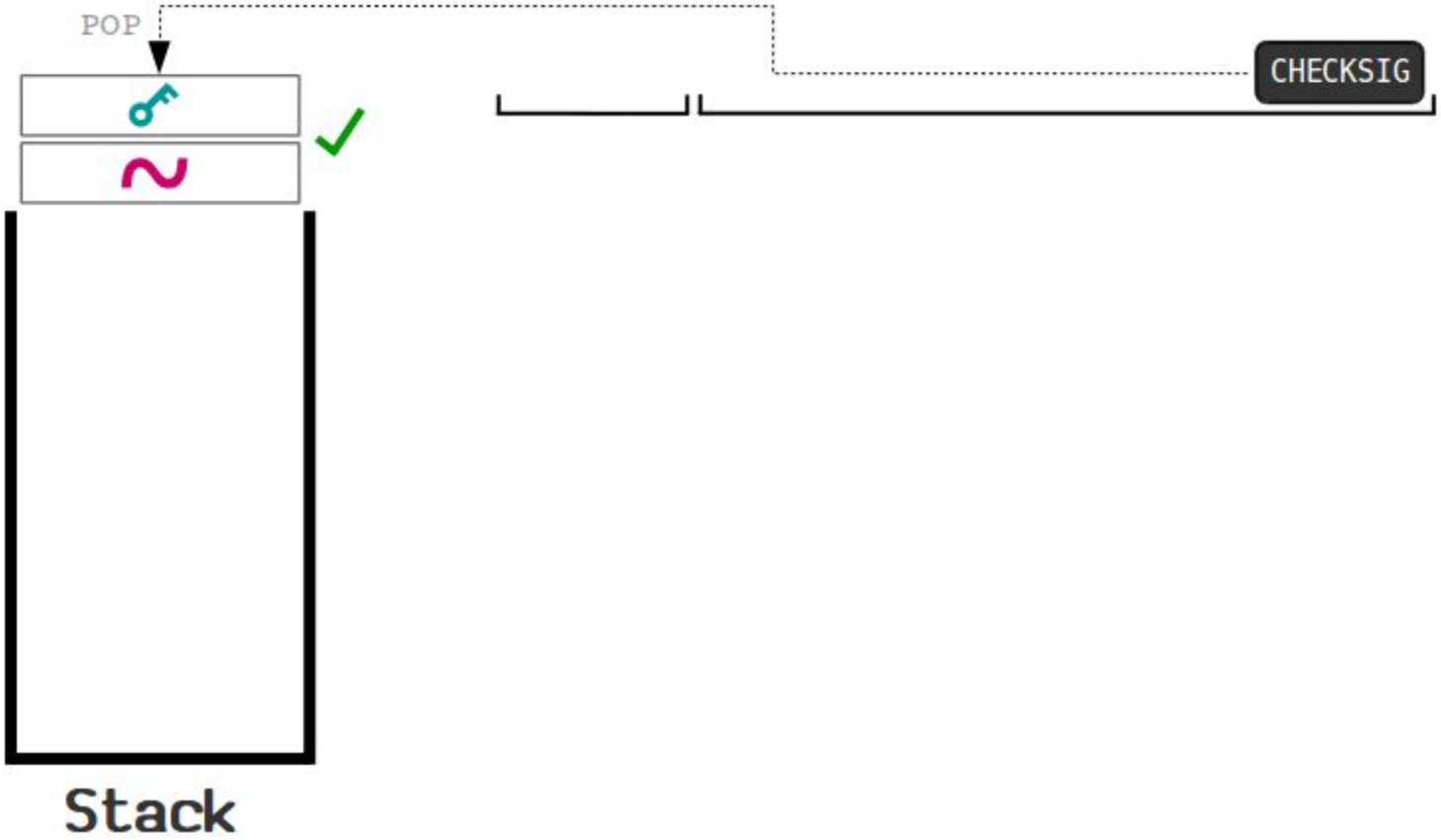

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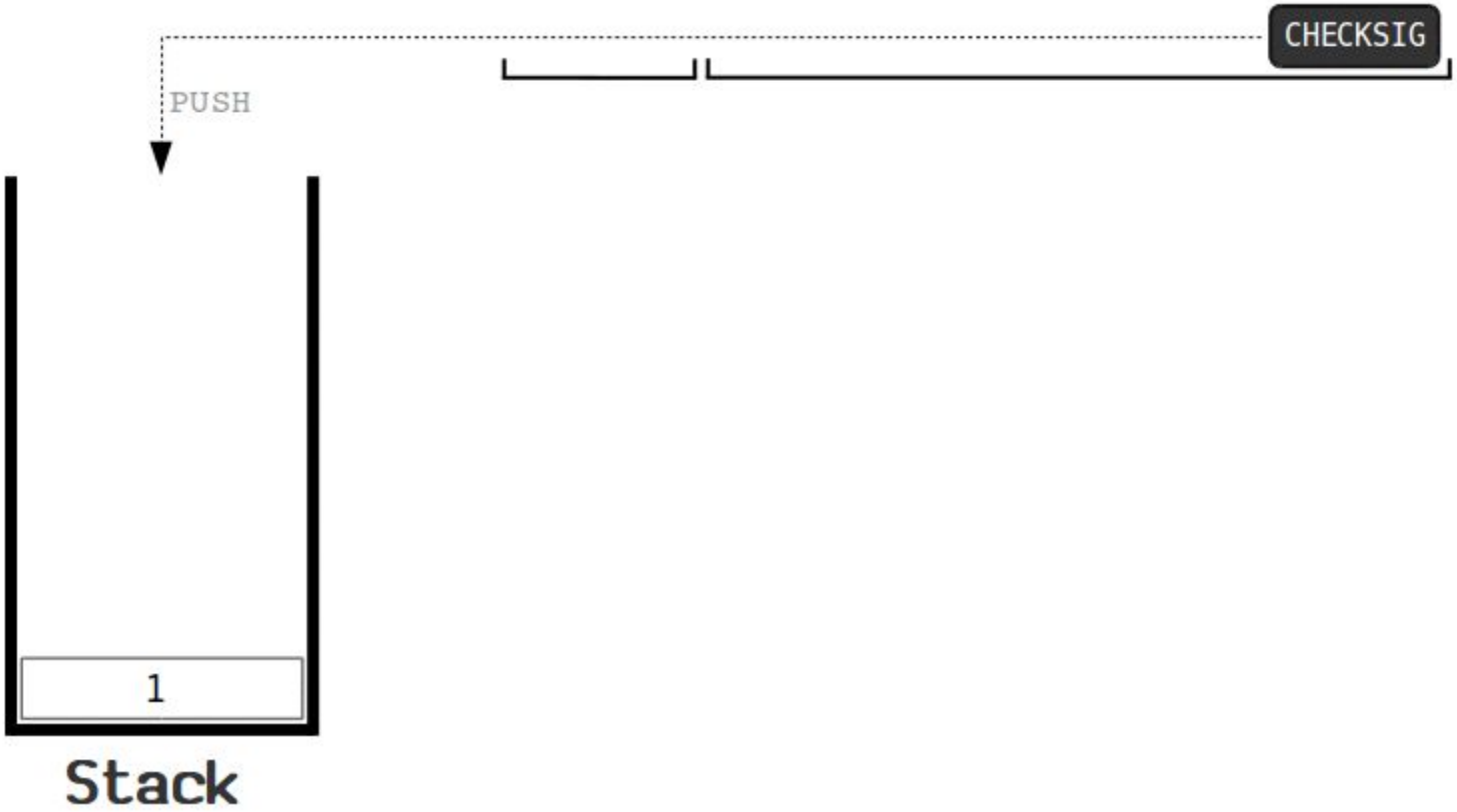
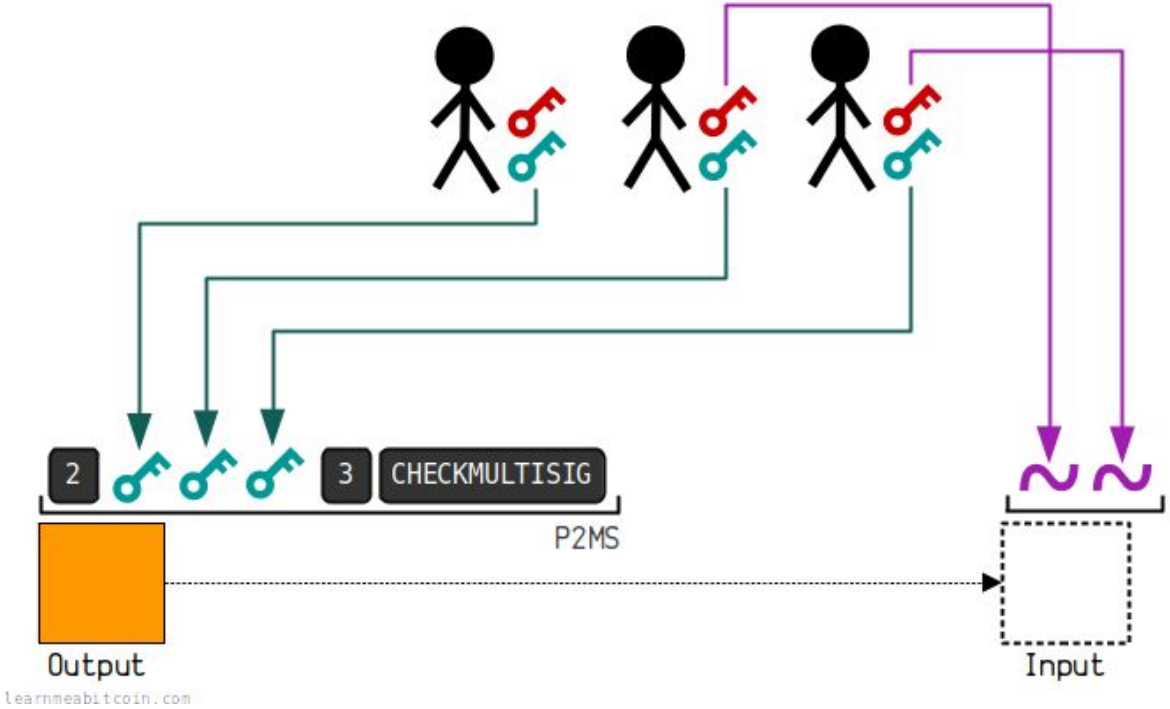
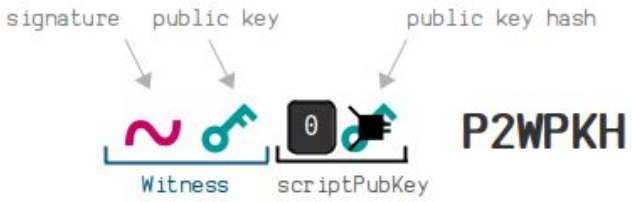
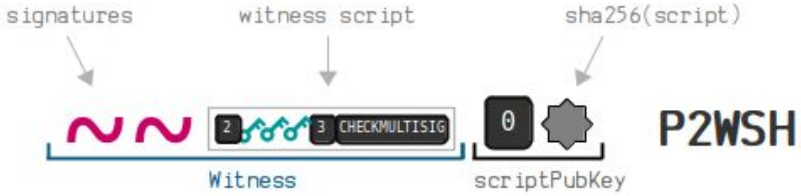


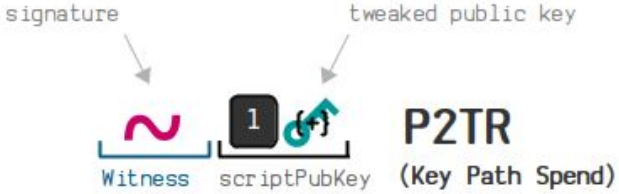
Multi Sig



Nuevos estándares







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Ejemplo

- Pizza day transaction
  https://www.blockchain.com/explorer/transactions/btc/a1075db55d416d3ca199f55b6084e2115b9345e16c5cf302fc80e9d5fbf5d48d
- Decoder
  https://paulkernfeld.com/bse/